Answer Key & Teacher Notes

Session 1 — Writing Algebraic Expressions · Session 2 — Evaluating Algebraic Expressions

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Writing Algebraic Expressions

TALK PROMPTS

- How do you decide which quantity in a story should become the variable?
- The $45 car-wash fee shows up only once. How does the expression show that it does not repeat?

PARTNER TASK

Concert Tickets and Two Items. One partner writes 8t + 25 for $8 tickets and a $25 fee and says what 8t means; the other writes 2j + 3f for juice at $2 a cup and muffins at $3 each. Switch and check each other's variable definitions.

WATCH FOR

- Putting the variable with the one-time fee (11 + 6b instead of 11b + 6).
- Adding instead of multiplying for a per-item rate (5 + d instead of 5d).
- Reversing "less than" (4 − y for "4 less than y").

SESSION 2 · TEACHER NOTES

Evaluating Algebraic Expressions

TALK PROMPTS

- Why must you substitute the number before you multiply or add? Explain the order.
- In 12g + 20, the gigabytes doubled from 3 to 6, but the cost did not. Why does the 20 keep that from happening?

PARTNER TASK

Team Points and Words to Symbols. One partner evaluates 3w + 2t for 5 wins and 4 ties (23); the other for 8 wins and 0 ties (24). Switch roles to write "8 less than n" (n − 8) and "n divided by 4, then increased by 3" (n ÷ 4 + 3).

WATCH FOR

- Writing digits side by side after substituting (4x with x = 3 read as 43).
- Adding before multiplying (4 + 2p with p = 5 as 6 × 5 = 30).
- Multiplying the coefficient before squaring ($5k^2$ with k = 3 as 15^2).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.A.2a	1	2	1	2	1	2	2	3	2	1	1
S1 · Form B	6.EE.A.2a	1	2	1	2	1	2	2	3	2	1	1
S2 · Form A	6.EE.A.2c	1	2	1	1	2	2	2	3	2	1	1
S2 · Form B	6.EE.A.2c	1	2	1	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

variable: A **letter** that stands for an unknown number.

coefficient: The number **multiplied** by the variable.

constant: A number by itself that does not **change**.

algebraic expression: An expression with at least one **variable**.

- An amount that repeats "each" time goes with the **variable**.
- A one-time amount is the **constant**.
- Order matters: "4 less than x" is x **–** 4.

Example: The $9 repeats each hour: 9 × *h* = **9h**. The one-time fee is the constant: **3**. Join the terms with a plus sign. Answer: The cost is **9h + 3** dollars.

YOU TRY

1. a. n + 6 b. 4k

2. 10t + 4

EXIT TICKET

1. a. y – 9 b. 7n

2. 15m + 20

3. **A** (8)

PRACTICE FORM A

1. a. n + 7 b. 5x c. y – 4 d. w ÷ 3 e. 2t + 1 f. 3k – 6

2. a. 6m; coefficient 6 b. 12; it is paid once, so it does not change with m c. 6m + 12

3. 8t + 25

4. 7c – 45

5. 17m + 40

6. Rani is wrong. $5 is paid for each day, so multiply: 5d. (5 + d adds only one $5.)

7. 3c + 2k

8. a. 25m + 40 b. 30m c. Gym A: 25 × 6 + 40 = $190; Gym B: 30 × 6 = $180. Gym B costs less for 6 months.

9. Bela put the variable with the one-time shipping fee. The $11 repeats for each book, so it goes with b: 11b + 6.

10. **A** (*n* + 7)

11. **B** (*p*)

Self-check word **SYMBOL**

PRACTICE FORM B

1. a. p + 9 b. 8z c. r – 10 d. b ÷ 6 e. 3s + 4 f. 5j – 2

2. a. 9m; coefficient 9 b. 16; it is paid once, so it does not change with m c. 9m + 16

3. 11t + 20

4. 6c – 35

5. 12m + 25

6. Tariq is wrong. $4 is paid for each hour, so multiply: 4h. (4 + h adds only one $4.)

7. 2j + 3f

8. a. 20m + 30 b. 24m c. Gym A: 20 × 5 + 30 = $130; Gym B: 24 × 5 = $120. Gym B costs less for 5 months.

9. Omar put the variable with the one-time shipping fee. The $9 repeats for each book, so it goes with k: 9k + 5.

10. **C** (5*x*)

11. **B** (*y* – 4)

Self-check word **LETTER**

GUIDED NOTES

substitute: **Replace** a variable with a given number.

evaluate: Find the **value** after you substitute.

variable: A letter that stands for a **number**.

- To evaluate, **substitute** the value for the variable.
- Then use the **order** of operations: powers, then × and ÷, then + and −.
- 4x means 4 × x. When x = 3, 4x = **12**, not 43.

Example: Substitute: 7 × **4** − 2. Multiply first: 7 × 4 = **28**. Subtract: 28 − 2 = **26**. Answer: 7*p* − 2 = **26** when *p* = 4

YOU TRY

1. a. 16 b. 5

2. 3 × 5 + 6 = 15 + 6 = $21

EXIT TICKET

1. a. 17 b. 26

2. 30 − 4 × 6 = 30 − 24 = 6 L

3. **A** (14)

PRACTICE FORM A

1. a. 33 b. 35 c. 25 d. 22 e. 40 f. 20

2. a. 4 × 3 + 5 = $17 b. 4 × 8 + 5 = $37 c. $37 − $17 = $20 d. The 5 is a flat fee added once to every cost, no matter how many hours.

3. 9 × 4 + 14 = 36 + 14 = $50

4. $5 \times 3^2 - 10 = 5 \times 9 - 10 = 45 - 10 = 35$

5. Yara added 4 + 2 first. 2p means 2 × p, so multiply first: 4 + 2 × 5 = 4 + 10 = 14.

6. a. 25 − 3 × 2 = 19 L b. 25 − 3 × 8 = 1 L c. As d grows, 3d grows, so less water is left.

7. a. 3 × 5 + 2 × 4 = 15 + 8 = 23 points b. 3 × 8 + 2 × 0 = 24 points

8. a. 12 × 3 + 20 = $56 b. 12 × 6 + 20 = $92 c. Only the 12g part doubles (36 to 72). The $20 fee stays the same, so the total does not double.

9. "Less than" flips the order: start with x, then subtract 15. Correct: x − 15; when x = 22, 22 − 15 = 7.

10. **A** (15)

11. **A** (number)

Self-check word **EVALUATE**

PRACTICE FORM B

1. a. 24 b. 27 c. 20 d. 54 e. 8 f. 18

2. a. 6 × 2 + 7 = $19 b. 6 × 5 + 7 = $37 c. $37 − $19 = $18 d. The 7 is a flat supply fee added once to every cost, no matter how many hours.

3. 8 × 6 + 15 = 48 + 15 = $63

4. $3 \times 4^2 + 7 = 3 \times 16 + 7 = 48 + 7 = 55$

5. Ken added 6 + 3 first. 3p means 3 × p, so multiply first: 6 + 3 × 4 = 6 + 12 = 18.

6. a. 12 + 5 × 3 = 27 L b. 12 + 5 × 9 = 57 L c. As h grows, 5h grows, so there is more water.

7. a. 3 × 6 + 2 × 2 = 18 + 4 = 22 points b. 3 × 7 + 2 × 0 = 21 points

8. a. 12 × 4 + 20 = $68 b. 12 × 8 + 20 = $116 c. Only the 12g part doubles (48 to 96). The $20 fee stays the same, so the total does not double.

9. "Less than" flips the order: start with y, then subtract 12. Correct: y − 12; when y = 30, 30 − 12 = 18.

10. **C** (30)

11. **B** (5)

Self-check word **VARIABLE**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · WRITING ALGEBRAIC EXPRESSIONS

1. a. n + 9 b. y − 8 c. 6k d. m ÷ 5 (or m/5) e. 2x + 3 f. 7(a + 2)

2. 15L + 20

3. a. p ÷ 4 b. 50 − s

4. 11p + 5; when p = 4: 11 × 4 + 5 = $49

5. He put the variable with the one-time fee. The $7 repeats for each ticket, so it goes with n: 7n + 3.

SHEET 2 · EVALUATING ALGEBRAIC EXPRESSIONS

1. a. 30 b. 23 c. 22 d. 40 e. 12 f. 10

2. a. 3 × 4 + 2 × 5 = 22 b. 16 − 5 = 11

3. $13, $29, $53

4. 100 − 6 × 5 = 100 − 30 = 70%

5. 8 (multiply first: 3 × 4 = 12, then 20 − 12 = 8)

6. **C** (8)

7. **D** (9*t* + 6)

October 2026 conceptual additions — teacher key

Session 1 · Writing Algebraic Expressions

7h + 4 dollars; 7h is hourly cost, 4 is charged once. It is an expression, not an equation, until an equality is stated.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 7h + 4 dollars; 7h is hourly cost, 4 is charged once. It is an expression, not an equation, until an equality is stated.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Evaluating Algebraic Expressions

6(3) + 4 = 18 + 4 = 22. For example, three $6 items plus one $4 fee.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 6(3) + 4 = 18 + 4 = 22. For example, three $6 items plus one $4 fee.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.